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ARTICLE INFORMATION

Article title

GYMED: GYM Exercise Dataset for Pose-based Action Classification

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Keywords

Gym exercise classification, pose estimation, dataset, human action recognition.

Abstract

Gym exercise is important in daily life as it can help improve physical and mental health. In scenarios where people have insufficient time to go to the center, doing gym exercises at home becomes the most suitable choice. However, without expert supervision, home exercisers often adopt incorrect form, leading to risking injury or, more seriously, unexpected results. Therefore, the demand for a virtual personal trainer, who can accurately recognize gym exercises and deliver correct feedback, has become pressing. To realize such a system, this dataset is built to provide data for deep learning models which serve as the core of gym exercise recognition tasks. The videos in the dataset are categorized into 22 classes of gym exercises. The dataset includes 580 videos for training, 208 for validation and 236 for testing, without any overlap of content between sets, for gym exercise classification using video. The videos are collected from multiple internet sources in two formats (.mp4 and .mov). The videos are collected in different camera angles, lighting conditions, subject appearances, and recording qualities, which indicates a deliberate strategy to enhance the dataset's diversity and robustness to real-world variations. In addition, action-relevant frames are manually segmented, which eliminates the redundant frames, such as transitions, rest periods or off-camera motion. This helps increase the fidelity of the dataset for gym exercise classification tasks. With the first publicly available numerous videos, this dataset is a valuable resource for deep learning models to detect and classify gym exercises using video.

SPECIFICATIONS TABLE

[Instruction for SPECIFICATIONS TABLE in comment box]

Subject	Computer Sciences
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Specific subject area	<i>Gym exercise classification, Video processing</i>
Type of data	<i>Video and Text Files</i>
Data collection	<i>The videos are collected from various public sources and the author's own recordings in different conditions of camera angles, lighting levels, subject appearances and recording qualities. Then the frames in each video are manually segmented to the ones containing the actions of a gym exercise before it is categorized into corresponding gym exercises.</i>
Data source location	<i>Workout/Exercise Video Dataset. YouTube fitness channels. Pexels Website [2]. Freepik Website [3]. recorded by the authors.</i>
Data accessibility	Repository name: The GYM Exercise Classification Dataset Data identification number: <i>(or DOI or persistent identifier)</i> Direct URL to data: https://www.kaggle.com/datasets/nguyenxuancuongk18dn/gym-exercise-classification-dataset
Related research article	<i>None</i>

VALUE OF THE DATA

- *This is the first large-scale, public video dataset of gym exercises. In addition, the videos that illustrate the actions of gym exercises in different conditions of camera angles, lighting levels can support the development of pose estimation and gym exercise classification algorithms.*
- *The frames in the videos in the dataset are segmented into the frames with and without actions, which can allow researchers to develop and validate the computer vision algorithms to detect the frames containing the action of gym exercises in videos.*
- *The dataset is a valuable resource for facilitating the development of practical applications of virtual personal trainers and automated exercise feedback systems.*

BACKGROUND

As general living standards rise, more individuals turn to gym workouts to improve both physical fitness and mental well-being. Without expert supervision, individuals doing exercises at home often adopt incorrect form, which can cause injury or suboptimal results. For such a scenario, the demand for a virtual personal trainer capable of accurately recognizing the performed exercise, and thereby delivering frequent feedback and correction, has become pressing. The increasing number of home exercisers without direct professional guidance directly drives the need for automated solutions. The core of such an automated system is AI-driven computer vision algorithms which can detect and recognize gym exercises. This suggests that video datasets of gym exercises have played an important role not only in developing AI models in academia but also in building practical applications that offer significant public health and fitness benefits. Although the previous datasets of gym exercises were published, the number of videos and classes was limited, which is suitable for small AI-based systems or academic research. Therefore, the videos collected from various internet resources in different conditions of lighting level, camera angles, subject appearances and recording qualities are the valuable resources for advancing deep learning approaches for gym exercise recognition.

DATA DESCRIPTION

Overall Dataset Summary

This report provides a summary of the GYM Exercise Classification Dataset statistics [4]:

- *Total videos: 1024*
- *Total capacity: 10225.37 MB*
- *Number of different resolutions: 33*
- *Number of different actions: 22*
- *Number of different file formats: 2*

List of 22 Gym Exercise Classes

The dataset includes 22 distinct gym exercises: barbell biceps curl, bench press, chest fly machine, deadlift, decline bench press, hammer curl, hip thrust, incline bench press, lat pulldown, lateral raise, leg extension, leg raises, plank, pull-up, push-up, Romanian deadlift, Russian twist, shoulder press, squat, t-bar row, tricep dips, and tricep pushdown. The selection of these 22 exercises represents a significant and diverse set of movements, ranging from isolated muscle groups to complex compound movements. This broad coverage increases the dataset's complexity and practical applicability, as models must learn to distinguish between subtle variations and distinct actions.

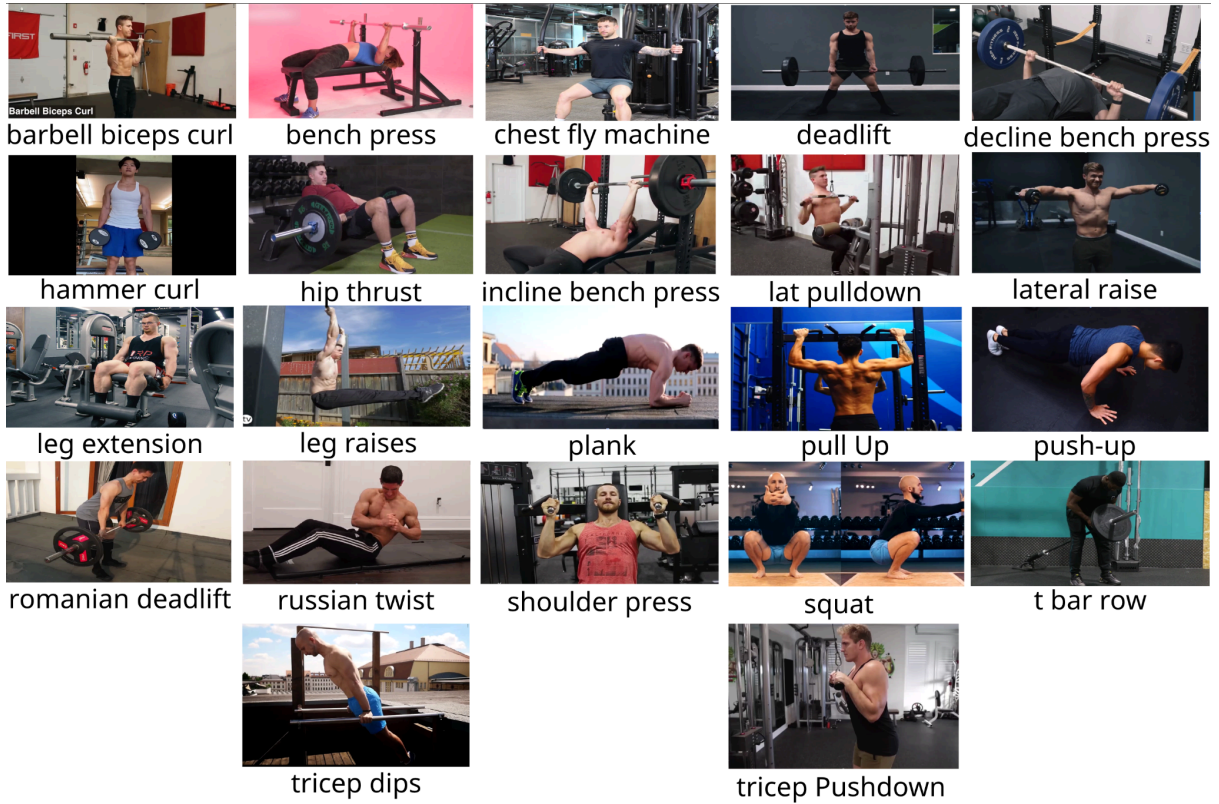


Figure 1: Frame samples from 22 classes of gym exercises in dataset

Detailed Analysis of Video Counts

The dataset contains a total of 1,024 unique videos. The dataset is split at the video level in a 6:2:2 ratio to prevent content overlap between the training, validation, and test sets.

Video-Level Splitting Strategy and Rationale

The dataset is split at the video level in a 6:2:2 ratio with the explicit aim of preventing content overlap between the training, validation, and test sets. This strategy is crucial to avoid data leakage, a phenomenon that can artificially inflate test performance and lead to an overestimation of a model's generalization capabilities to unseen data. Splitting at the video level ensures that a specific video and all segments/clips generated from it reside in only one dataset split, preventing models from memorizing video-specific patterns rather than learning general exercise characteristics.

File Format Statistics

The distribution of video file formats in the dataset is as follows:

Table I: Distribution of Video File Formats

File Format	Number of Videos	Percentage (%)
.mp4	964	94.1
.mov	60	5.9

Resolution Statistics

The dataset comprises videos with 33 different resolutions. A summarized breakdown by resolution is provided below, focusing on resolutions with more than 20 videos, with others grouped for conciseness:

Table II: Summarized Video Counts per Resolution

<i>Resolution</i>	<i>Number of Videos</i>
1280x720	385
1920x1080	315
720x1280	130
854x480	35
640x360	28
3840x2160	23
Resolutions with ≤ 20 videos (27 others)	108

The resolutions with 20 or fewer videos collectively account for 108 videos. These resolutions vary in total capacity, associated actions, dataset splits (train, val, test), file formats (.mp4, .mov), and were processed using ffprobe.

Class Imbalance Analysis

The distribution of videos across the 22 gym exercise classes reveals significant imbalance. The video counts per action are presented in Table III (with percentages based on total 1024 videos).

Table III: Video counts per exercise class and their percentage

<i>Class</i>	<i>Number of Videos</i>	<i>Percentage (%)</i>
barbell biceps curl	84	8.2
bench press	84	8.2
lat pulldown	71	6.9
push-up	66	6.4
tricep Pushdown	65	6.3
pull Up	52	5.1
deadlift	50	4.9
shoulder press	50	4.9
squat	49	4.8
lateral raise	47	4.6
hammer curl	46	4.5
incline bench press	46	4.5
chest fly machine	45	4.4
leg extension	44	4.3

<i>tricep dips</i>	43	4.2
<i>leg raises</i>	32	3.1
<i>decline bench press</i>	32	3.1
<i>hip thrust</i>	29	2.8
<i>t bar row</i>	29	2.8
<i>russian twist</i>	27	2.6
<i>romanian deadlift</i>	24	2.3
<i>plank</i>	9	0.9

This pronounced imbalance—for instance, "plank" is represented by only 9 videos (0.9%) while "barbell biceps curl" has 84 videos (8.2%) and "bench press" has 84 videos (8.2%)—can bias the classifier toward well-represented classes and severely degrade performance on underrepresented ones unless addressed by augmentation or sampling strategies. The detailed presentation of class imbalance is a crucial transparency point for the dataset. This sets a clear direction for future research using this dataset, emphasizing the need for specialized imbalance-handling techniques.

Each raw video is manually trimmed into "clean" segments to contain only the target exercise. This precise segmentation was achieved by referring to pre-generated `.txt` files, manually created by the authors. These `.txt` files meticulously define the start and end timestamps of relevant exercise actions within each video, thereby effectively excluding transition scenes, irrelevant footage, setup sequences, advertisements, and conversations. This meticulous process ensures that each segment contains only the intended exercise, maximizing data quality for classification. On average, each video yielded 1.09 segments, producing a total of 1,108 segments across the entire dataset: 639 in the training set, 210 in validation, and 259 in the test. This segmented dataset, along with its metadata, is available as part of the public Kaggle dataset [4]. The detailed distribution of segments per exercise class by split is provided in Table IV.

Table IV: Video and segment counts per exercise class by split

Class	Videos			Segments		
	Train	Val	Test	Train	Val	Test
<i>barbell biceps curl</i>	52	18	14	53	18	14
<i>bench press</i>	51	18	15	51	18	15
<i>lat pulldown</i>	28	9	8	29	9	8
<i>push-up</i>	30	10	10	30	10	10
<i>tricep Pushdown</i>	18	5	9	29	5	10
<i>pull Up</i>	24	8	14	28	8	14
<i>deadlift</i>	14	6	9	19	6	15
<i>shoulder press</i>	27	10	9	27	10	9
<i>squat</i>	42	15	14	42	15	15
<i>lateral raise</i>	24	8	15	24	8	15
<i>hammer curl</i>	22	9	13	27	9	13
<i>incline bench press</i>	15	6	11	15	6	11
<i>chest fly machine</i>	5	2	2	5	2	2
<i>leg extension</i>	31	11	10	31	11	10
<i>tricep dips</i>	40	14	12	43	14	14

<i>leg raises</i>	12	6	6	13	6	8
<i>decline bench press</i>	15	6	6	19	6	6
<i>hip thrust</i>	27	10	13	28	10	14
<i>t bar row</i>	24	10	15	27	11	17
<i>russian twist</i>	15	4	10	17	4	10
<i>romanian deadlift</i>	39	14	12	39	14	12
<i>plank</i>	25	9	9	43	10	17

EXPERIMENTAL DESIGN, MATERIALS AND METHODS

Data collection

Primary data were sourced from public datasets or from platforms, including Pexels, Freepik, and YouTube Fitness. For each class, 1-2 high-quality videos with clear, distinct movements were collected from each source. These raw videos were then trimmed into specific segments, strictly isolating the target exercises and eliminating any irrelevant frames or extraneous actions. Specifically, 652 videos were recovered from a public Workout/Exercise Video Dataset, 103 clips were downloaded from YouTube fitness channels, 65 videos were recovered from Pexels Website [2], 76 videos were recovered from Freepik Website [3].

To expand and enrich the dataset, 128 supplementary videos were collected at a fully equipped gymnasium for all 22 gym exercise classes. Filming was conducted using two human models, with each video lasting between 20 to 30 seconds. To enhance dataset diversity, each exercise was recorded using two techniques: a fixed camera angle and a dynamic, moving camera to capture the full range of motion from multiple perspectives.

Data segmentation

After collection, the entire video is analyzed to precisely extract the parts that can perform the targeted exercise. Each video is cut into short segments corresponding to the completion of the task once, ensuring the removal of irrelevant parts such as rest times, scene changes, or movements outside the exercise. The analysis process is performed manually with the support of video processing tools to ensure high accuracy.

Cross verification

A rigorous data quality control process was implemented. Each team member was independently responsible for preprocessing a subset of videos, evaluating the appropriateness of duration, camera angles, and visual clarity. Subsequently, a mutual cross-check was performed among team members to verify accuracy, ensuring all data was correctly and precisely categorized into the 22 predefined classes prior to model training.

LIMITATIONS

Reiteration of Class Imbalance and Its Impact

The dataset includes some underrepresented classes like “plank”, which causes imbalance in the dataset. This imbalance can lead to models being biased towards majority classes, potentially limiting their real-world applicability where all exercises need to be reliably recognized.

Exercises with subtle movement differences or persistent data scarcity are likely to pose challenges. Conversely, highly distinct movements (e.g., push-up, leg extension) are expected to be robustly recognized by deep learning models.

ETHICS STATEMENT

The work described in this article involves human subjects and data collected from social media and public platforms.

For the 128 videos recorded by the authors: Informed consent was obtained from all participants appearing in the videos. The data collection protocol was carried out in accordance with the Declaration of Helsinki and followed ethical standards.

For the data collected from online platforms (Pexels, Freepik, YouTube, and Kaggle): The data was collected in full compliance with the Terms of Service and redistribution policies of each respective platform. The videos from Pexels and Freepik are used under their specific licenses which allow for such usage. Data collected from YouTube and Kaggle consists of publicly available content, and no private or personally identifiable information beyond what is publicly displayed was harvested.

CRediT AUTHOR STATEMENT

Nguyen Xuan Cuong: Conceptualization, Methodology, Writing - Original draft preparation.

Truong Nhat Quang: Resources, Data curation, Validation, Visualization.

Vo Quoc Trinh: Conceptualization, Review & Editing, Supervision.

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DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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